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Paper Code : CE(ES)301/ES-AUE301/ES-ME301 Engineering Mechanics

UPID : 003490

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) What is virtual work?

(ii) What is meant by impulse momentum?

(iii) What is the advantage of D Alembert's principle?

(iv) What are the types of forced vibration?

(v) What is the unit of couple?

(vi) What is dynamic friction?

(vii) Two forces of 100 N and 150 N are acting simultaneously at a point. What is the resultant of these two forces, if the angle between them is 45°?

(viii) What is centroid?

(ix) What is the principle of transmissibility of forces?

(x) What is redundant frame?

(xi) What is trajectory?

(xii) A bullet is fired with a velocity of 100 m/s at an angle of 45° with the horizontal. How high the bullet will rise ?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. State D'Alembert's Principle. [5]

Kinetic energy depends on the distribution of mass and angular speed. Explain.

3. What are free, forced and damped vibrations? [5]

4. State and prove Lami's theorem. [5]

5. The resultant of the two forces, when they act at an angle of 60^0 is 14 N. If the same forces are acting at right angles, their resultant is $\sqrt{137}$ N. Determine the magnitude of the two forces. [5]

6. Prove that the angle of friction (ϕ) is equal to the angle made by an inclined plane with the horizontal when a solid body, placed on the inclined plane, is about to slide down. [5]

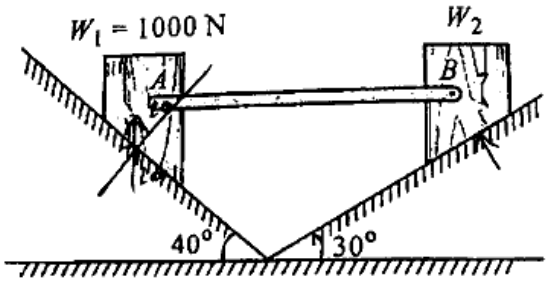
Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) A particle is projected in air with a velocity of 100 m/s and at an angle of ~~300~~ 30° with the horizontal. Find- i) The horizontal range ii) The maximum height by the particle iii) The time of flight. [8]

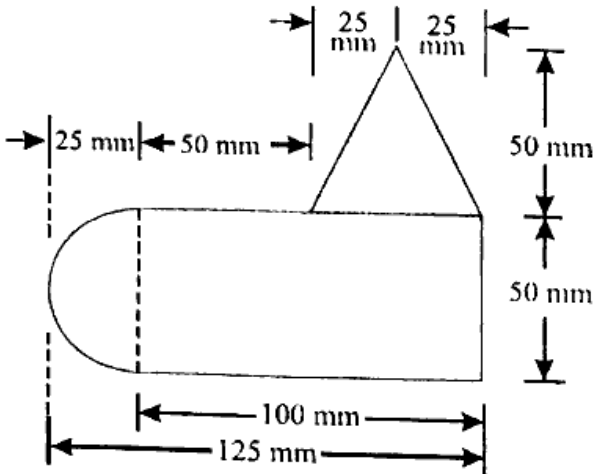
(b) A particle is moving with constant acceleration a. It covers initial distance of 16 m in 10 sec. What time it will take to cover entire distance of 400 m? What will be its final velocity? Assumed that the particle has start from rest. [7]

8. Two blocks W1 and W2 resting on two inclined planes are connected by a horizontal bar AB as shown in Fig. If W1 is equals 1000 N, determine the maximum value of W2 for which the equilibrium can exists. The angle of limiting friction is 20^0 at all rubbing faces. [15]

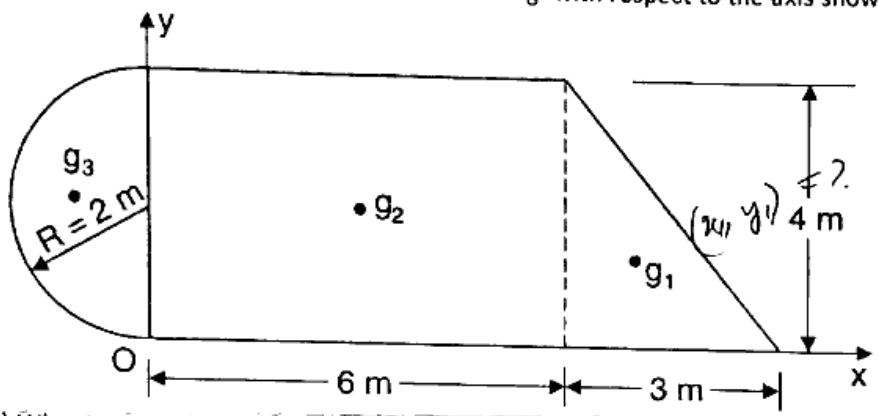


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9. (a) A uniform lamina shown in Fig. consists of a rectangle, a circle and a triangle. Determine the center of gravity of the lamina. All dimensions are in mm. [8]

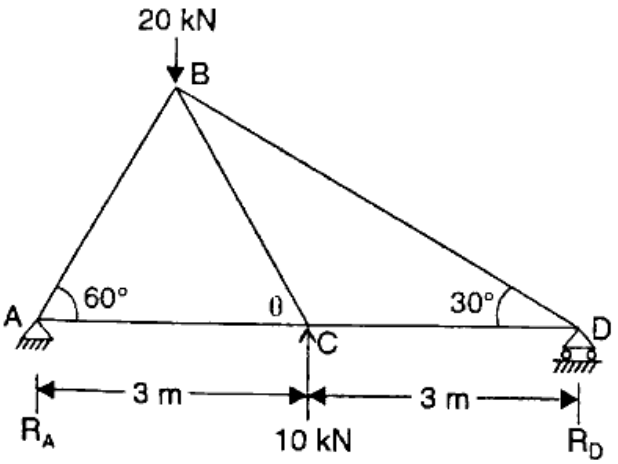


(b) Determine the centroid of the area shown in Fig. with respect to the axis shown. [7]



10. (a) i) What is a cantilever truss? How will you find out its reactions? [5]
ii) State the assumptions made in the analysis of pin jointed trusses.

(b) Determine the forces in all the members of the truss shown in Fig. [10]



11. (a) A railway wagon weighs 0.25 MN when empty and 0.6 MN when loaded. The spring of the wagon get compressed by 8 cm when it is full. Determine the natural period of vibration when the wagon is empty and when loaded. [7]

(b) Write down the causes and effects of vibrations in machines. [8]

*** END OF PAPER ***

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